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B28: SQL SuperHero Program : Session 12

1 message

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Write an SQL query to find employees who earn above the average salary of their respective department.
1. I want to find employees who are earning more than their department average
2. I want employee salary and i want department average salary
3. once i get it, i will compare and see where employee salary is more than department salary

```
with n1 as (
select department_id, avg(salary) dept_avg_salary from hr.employees group by department_id
), N2 as ( select E.Employee_Id, E.First_name, E.Last_name, E.salary, N1.dept_avg_salary
from N1, hr.employees E
Where N1.department_id = E.department_id
  ) select * from N2 Where SALARY > dept_avg_salary;
```

Check the first N1 block
1. select department_id, avg(salary) dept_avg_salary from hr.employees group by department_id
--> It gives us department_id and average salary of that department.
Our requirement is to get all such employees whose salary is more than department average. Employee salary is already in the employee table and for department average salary, we wrote our query. Now, consider we have 2 tables,
N1: Department and Dept_Avg_salary
Employee: It has employee data along with in which department they work
2. Select E.Employee_Id, E.First_name, E.Last_name, E.salary, N1.dept_avg_salary
from N1, hr.employees E
Where N1.department_id = E.department_id

In This N2 block, will join N1 with the Employees table and get the respective salary. Now compare and get SALARY > dept_avg_salary

==

Team 1	Team 2	Result
A	B	A
A	C	C
B	D	D
A	D	A

Output expected as :

Team	Won	Loss
A	2	1
B	0	~~1~~ 2
C	1	0
D	1	1

Lets Understand this sql
Question is : To give me result of how many teams played, how many win and how many loss.
very simple English. database is pure english, lets break this scenario to simple english language
1. Find out How many times team are playing,
2. Which team won how many matches
3. minus it from total match payed and we got how many they loose

1. select team1 as team from match
   union all
   select team2 as team from match;

Simple UNION ALL will let u combine, both team1 and team2 in table and we able to find how many times each team played

```
With N2 as (
select team, count(*) total_match from (
    select team1 as team from match
    union all
    select team2 as team from match
    ) Group By team
) select * from n2
```

output

TEAM	TOTAL_MATCH
B	2
C	1
A	3
D	2

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4 rows selected.

simple 1 line query and you got that which team is played how many times. simple straight fwd
Now by using with clause, we create a memory with this output. Database use this output as a table and do next calculation. Our next calculation to find how many they win. That is another column. we have Result column, which is showing that who won
lets find out who won the match

```
select result team, count(*) win from match group by result;
```

TEAM	WIN
C	1
A	2
D	1

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who win how many times

Now see, our first one line query tell us that who played how many matches and second 1 line query told us who win how many matches
Lets combine both small small query. With clause enable you that write small small code and combine them all. 1 sql with multiple select statements

```
 With N2 as (
select team, count(*) total_match from (
   select team1 as team from match
   union all
   select team2 as team from match
   ) Group By team
) select * from n2,
N3 as (
select result team, count(*) win from match group by result
),
N4 as (
select n2.team, n2.total_match, nvl(n3.win,0) win
from n2 left join n3 on n2.team = n3.team
) select * from N4;
```

```
With N2 as (
select team, count(*) total_match from (
    select team1 as team from match
    union all
    select team2 as team from match
    ) Group By team
),
N3 as (
select result team, count(*) win from match group by result
),
N4 as (
select n2.team, n2.total_match, nvl(n3.win,0) win
from n2 left join n3 on n2.team = n3.team
) select * from N4
```

1st Part we wrote

2nd Part we wrote

Lets combine

How many they played

TEAM	TOTAL_MATCH	WIN
C	1	1
A	3	2
D	2	1
B	2	0

How many they win

Download CSV

4 rows selected.

but we have to find out how many we loss. lets minus total_match - win. adding this line

select N4.Team, N4.win, N4.total_match - n4.win as loss from n4

```
 With N2 as (
select team, count(*) total_match from (
   select team1 as team from match
   union all
   select team2 as team from match
   ) Group By team
),
N3 as (
select result team, count(*) win from match group by result
),
N4 as (
select n2.team, n2.total_match, nvl(n3.win,0) win
from n2 left join n3 on n2.team = n3.team
) select N4.Team, N4.win, N4.total_match - n4.win as loss from n4;
```

```sql
With N2 as (
select team, count(*) total_match from (
    select team1 as team from match
    union all
    select team2 as team from match
    ) Group By team
),
N3 as (
select result team, count(*) win from match group by result
),
N4 as (
select n2.team, n2.total_match, nvl(n3.win,0) win
from n2 left join n3 on n2.team = n3.team
) select N4.Team, N4.win, N4.total_match - n4.win as loss from n4
```

TEAM	WIN	LOSS
C	1	0
A	2	1
D	1	1
B	0	2

Download CSV

4 rows selected.

output

If you do not know how to start, use WITH

==
==================================

SELF JOIN

A self join is a join in which a table is joined with itself (which is also called Unary relationships). In a self-join, each row of the table is joined with itself and all the other rows of the same table. Thus, a self-join is mainly used to combine and compare the rows of the same table in the database.

Most common question on Self Join is:

Find out the employee_name and manager_name from employee_table

```sql
SELECT
   e.employee_id, e.first_name employee_name,
   m.employee_id manager_id, m.first_name  maanger_name
FROM
   hr.employees e
INNER JOIN hr.employees m ON m.employee_id = e.manager_id
```

If you take a look at this query, we have one table that has employee_id and manager_id both and every manager is an employee. We have to find both the names. so here same table will be used 2 times. **Hr.employee as E and Hr.employee as M**

I gave 2 different aliases to the same table and used them 2 times. One time called them E and other time call them M. Same table as E, use to select employee and same table as M, use to pick manager.

```sql
SELECT
    e.employee_id, e.first_name employee_name,
    m.employee_id manager_id, m.first_name  maanger_name
FROM
    hr.employees e
INNER JOIN hr.employees m ON m.employee_id = e.manager_id
```

SELF

EMPLOYEE_ID	EMPLOYEE_NAME	MANAGER_ID	MAANGER_NAME
168	Lisa	148	Gerald
169	Harrison	148	Gerald
170	Tayler	148	Gerald
171	William	148	Gerald
172	Elizabeth	148	Gerald
173	Sundita	148	Gerald
103	Alexander	102	Lex
162	Clara	147	Alberto

with n1 as (
select employee_id, first_name employee_name, manager_id from hr.employees)
select n1.*, E.first_name manager_name
from n1, hr.employees E
where n1.manager_id = E.employee_id

```sql
with n1 as (
    select employee_id, first_name employee_name, manager_id from hr.employees)
select n1.*, E.first_name manager_name
from n1, hr.employees E
where n1.manager_id = E.employee_id
```

1 gives me all Emp & their manger

2 Here i join them again with Emp table & get manager name

EMPLOYEE_ID	EMPLOYEE_NAME	MANAGER_ID	MANAGER_NAME
168	Lisa	148	Gerald
169	Harrison	148	Gerald
170	Tayler	148	Gerald

SQL query to find Employees earning more than their mangers

Self Join:
understand this, here we first need to find out that every employee and their managers, once we know the employee and their managers then we knew salary as well and compare that salary

SELECT e1.EMPLOYEE_ID emp, e2.EMPLOYEE_ID as Manager
FROM hr.Employees e1, hr.Employees e2 where e1.Manager_Id = e2.EMPLOYEE_ID;

```
1  SELECT e1.EMPLOYEE_ID emp, e2.EMPLOYEE_ID as Manager
2  FROM hr.Employees e1, hr.Employees e2 where e1.Manager_Id = e2.EMPLOYEE_ID
3
4
```

Same table = self join

EMP	MANAGER
101	100
102	100
103	102
104	103
105	103
106	103
107	103
108	101
109	108

Now we just add a check of salary.
and e1.salary > e2.salary

SELECT e1.EMPLOYEE_ID emp, e2.EMPLOYEE_ID as Manager
FROM hr.Employees e1, hr.Employees e2 where e1.Manager_Id = e2.EMPLOYEE_ID
and e1.salary > e2.salary

```
select E.Employee_Id, E.First_name emp_name, E.salary emp_salary, m.employee_id maanger_id, m.first_name manager_name, m.salary manager_salary
    from hr.employees E inner join hr.employees M
on E.manager_id = M.Employee_id
    where E.salary > m.salary
```

Every manager is employee

Salary of Emp

EMPLOYEE_ID	EMP_NAME	EMP_SALARY	MAANGER_ID	MANAGER_NAME	MANAGER_SALARY
168	Lisa	11500	148	Gerald	11000
174	Ellen	11000	149	Eleni	10500

2 rows selected.

With clause

```
with n1 as
   (select employee_id,salary emp_sal,manager_id from hr.employees),
   N2 as(
   select n1.*,emp.salary as MGR_SAL from n1,hr.employees emp
   where n1.manager_id=emp.EMPLOYEE_ID)
   select * from n2 where EMP_SAL>MGR_SAL
```

```
with n1 as
    (select employee_id,salary emp_sal,manager_id from hr.employees)
    N2 as(
    select n1.*,emp.salary as MGR_SAL from n1,hr.employees emp
    where n1.manager_id=emp.EMPLOYEE_ID)
    select * from n2 where EMP_SAL>MGR_SAL
```

EMPLOYEE_ID	EMP_SAL	MANAGER_ID	MGR_SAL
168	11500	148	11000
174	11000	149	10500

2 rows selected.

First we get in N1 Block, The employee_id, name and manager_id . These 3 columns will be straightforward to get from the employee table. Now after N1 block, my output has 3 columns
Employee_Id, Employee_Salary , Manager_id
Now, What do we want? Salary of that Manager_id. If we have it then we can directly compare. Just think that every manager is an employee, so we have to check the salary of that manager_id.

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Question: I have a table that has all the country name who will play against each other. Write a sql to find the maximum number of unique matches possible.
For example India VS Pakistan is 1 match. Pakistan Vs India will not consider unique.

```
create table match (team varchar2(20));

insert into match values ('India');
insert into match values ('Pakistan');
insert into match values ('Australia');
insert into match values ('England');
insert into match values ('South Africa');
insert into match values ('West Indies');
```

Here we can see both team will come from same table. So it means self Join.

```
select A.team , B.Team
from match A, Match B
where A.team > B.Team;
```

```
select A.team, B.Team
    from match A, match B
where A.team > B.team
```

All combination

choose 1